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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

Department of CSE

B.Tech. CSE Trimester -5

Roll Number
2520030469
Name
Mahitha Reddy

Cloud Infrastructure and Services
25CS2206E

LAB RECORD

2026

Experiment -1

AIM: Create an IAM user with programmatic access only, download the access key and secret key, and document the steps with screenshots.Then set up a billing alert for the account so that an email is received when the Free Tier limit is about to be crossed.

Description:

AWS Identity and Access Management is a service that allows users to securely manage access to AWS resources. An IAM user with programmatic access can interact with AWS services using the AWS CLI, SDKs, or APIs without requiring AWS Management Console access.

AWS provides access credentials consisting of an Access Key ID and a Secret Access Key for authentication.

AWS Billing and Cost Management allows users to monitor their cloud expenses. By enabling Free Tier usage alerts and configuring AWS Budgets, users can receive email notifications when their usage approaches specified limits, helping prevent unexpected charges.

Steps:

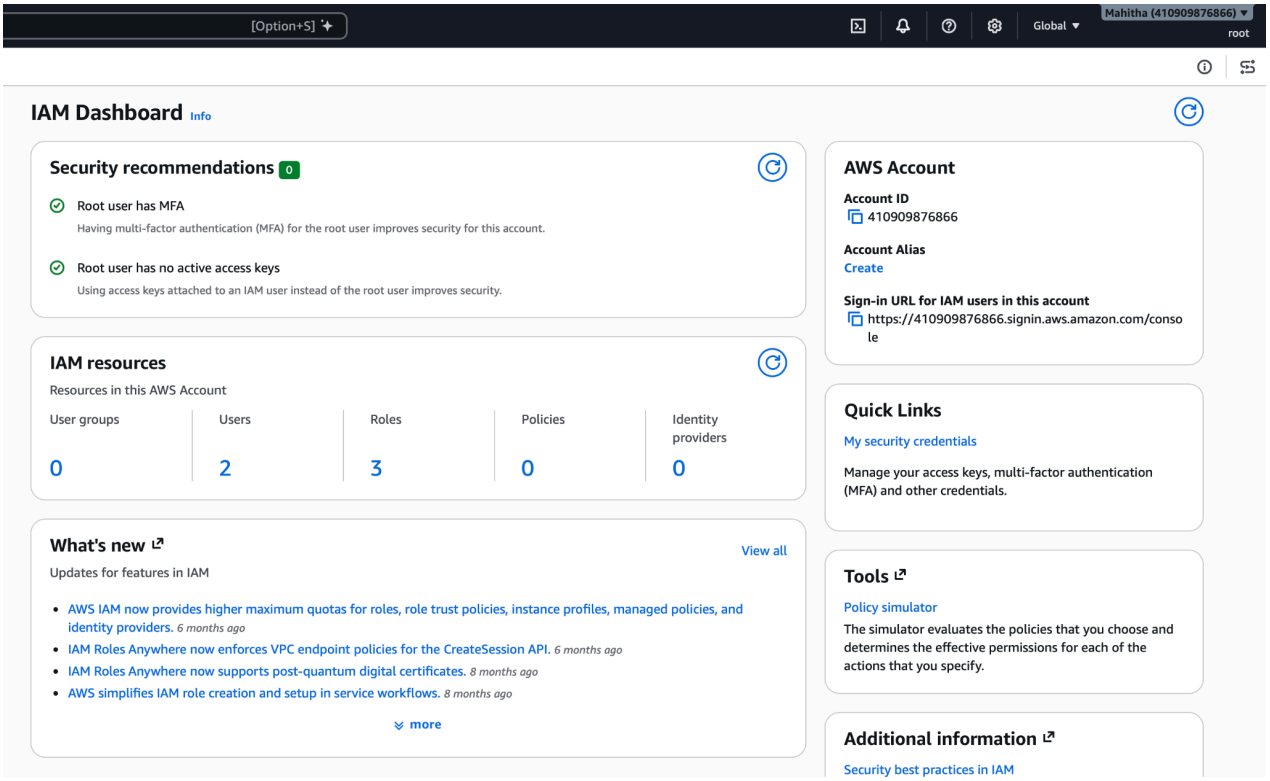
IAM User Creation

1. Log in to AWS Console.
2. Open IAM → Users → Create user.
3. Enter username and disable console access.
4. Assign permissions and create the user.
5. Open Security credentials → Create access key.
6. Generate and download the keys securely.

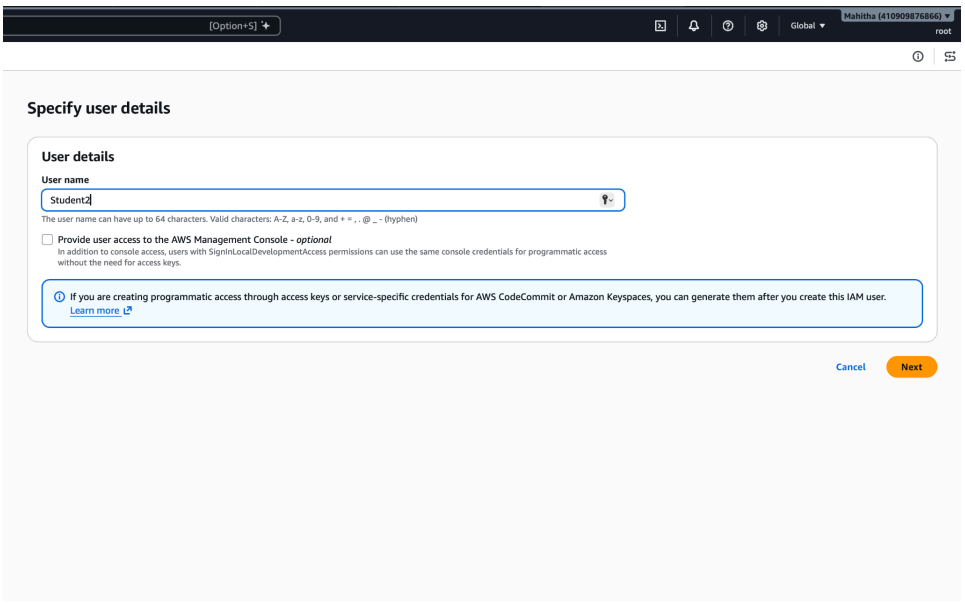
Zero-Spend Budget Alert

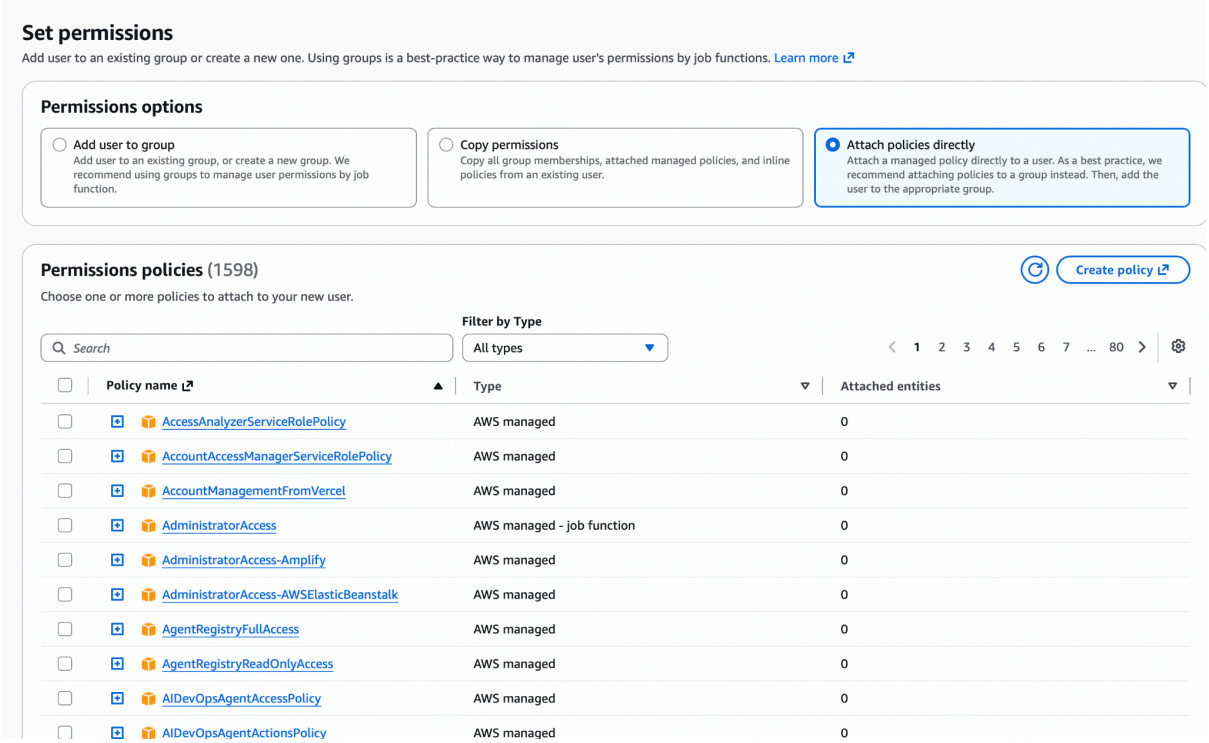
1. Open Billing and Cost Management.
2. Go to Budgets → Create budget.
3. Select Zero spend budget.
4. Enter your email address.
5. Click Create budget.

Output:

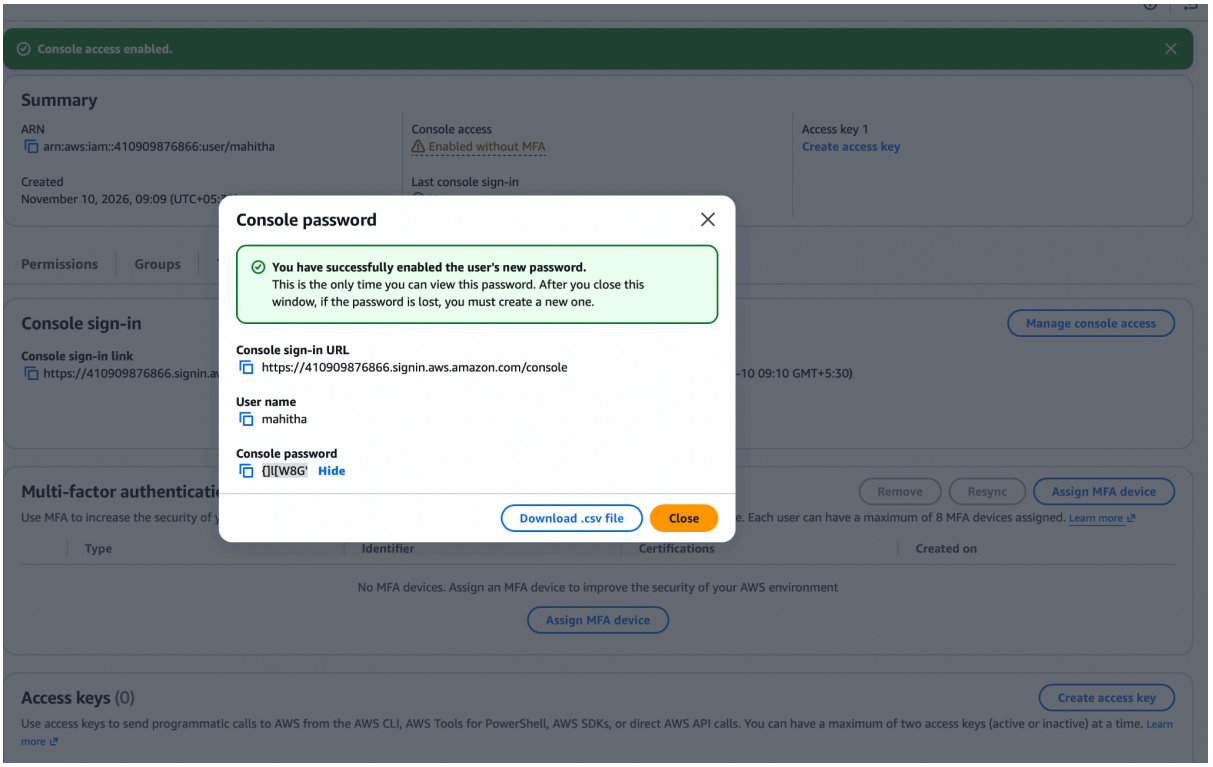


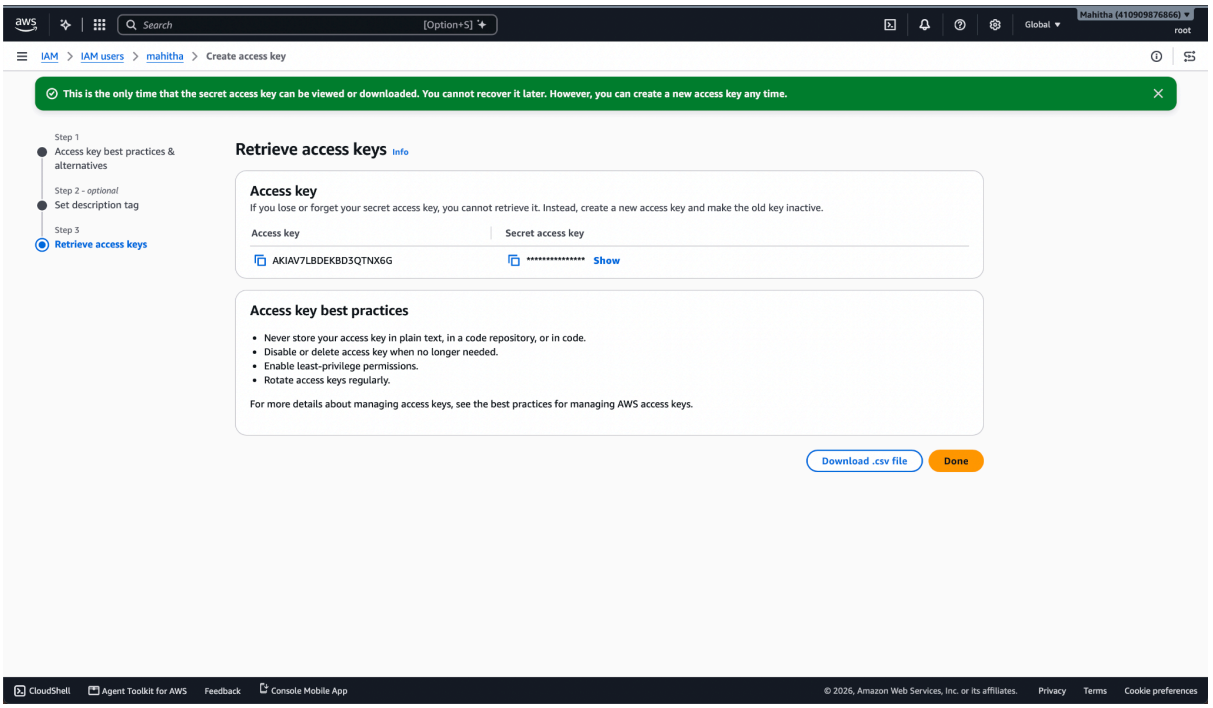
Dashboard

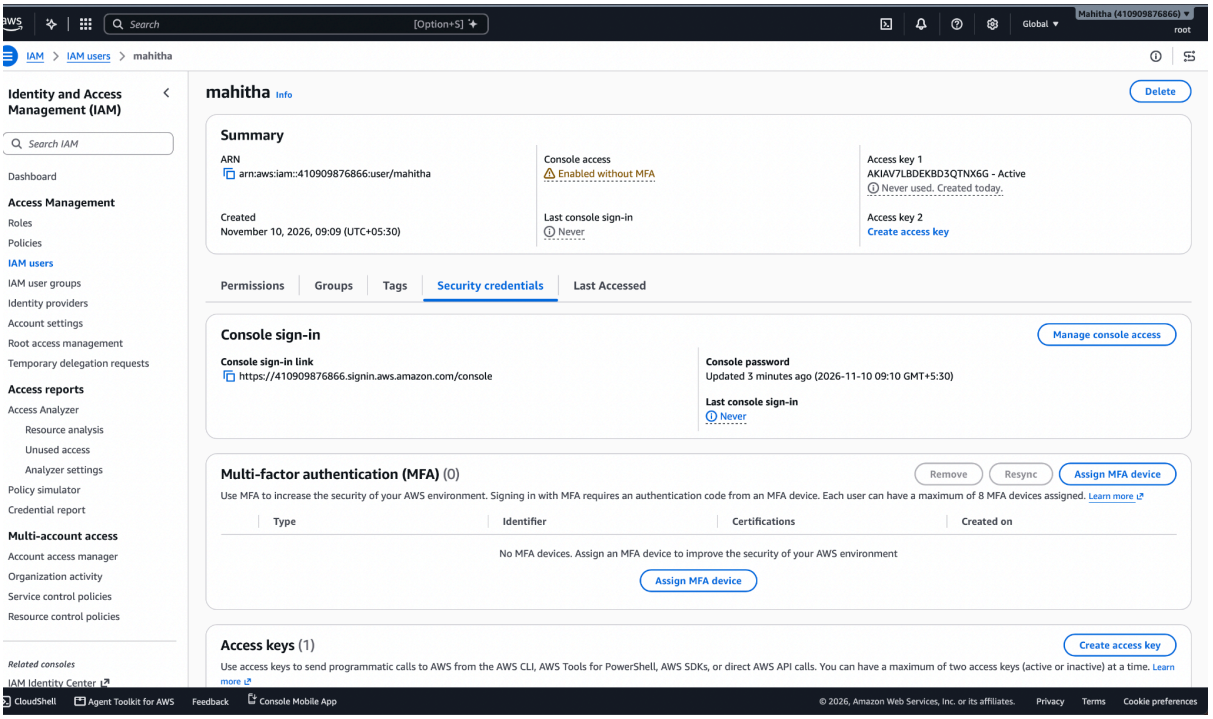


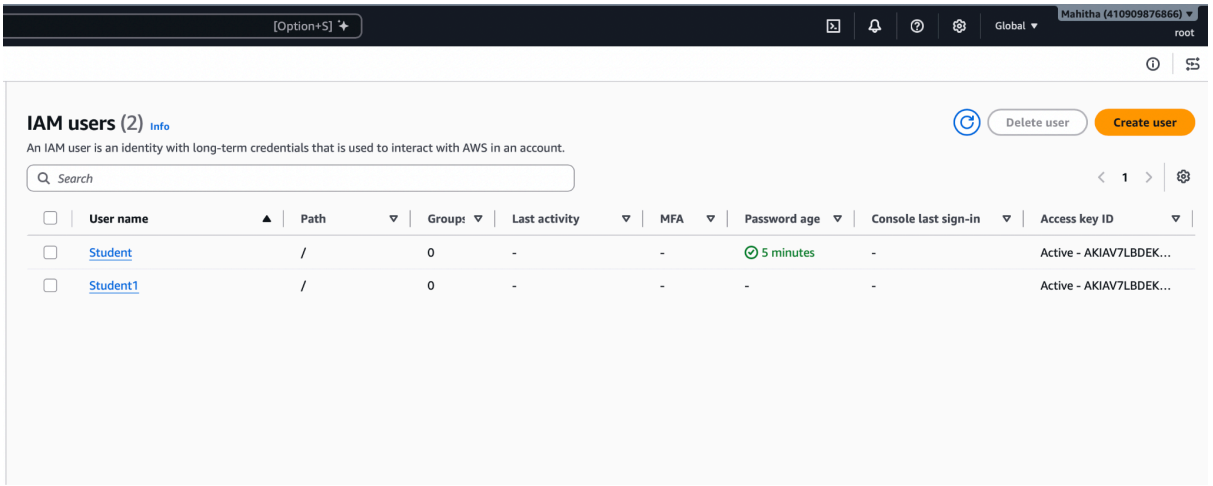


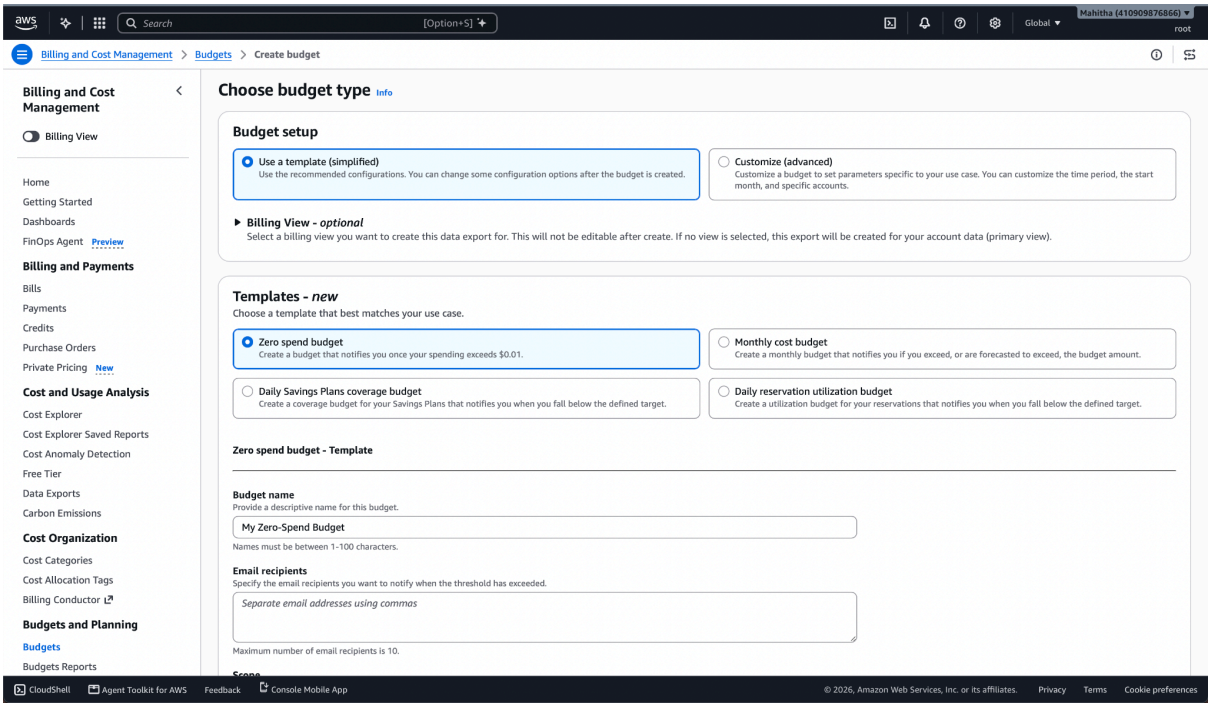
Creating user and giving permissions

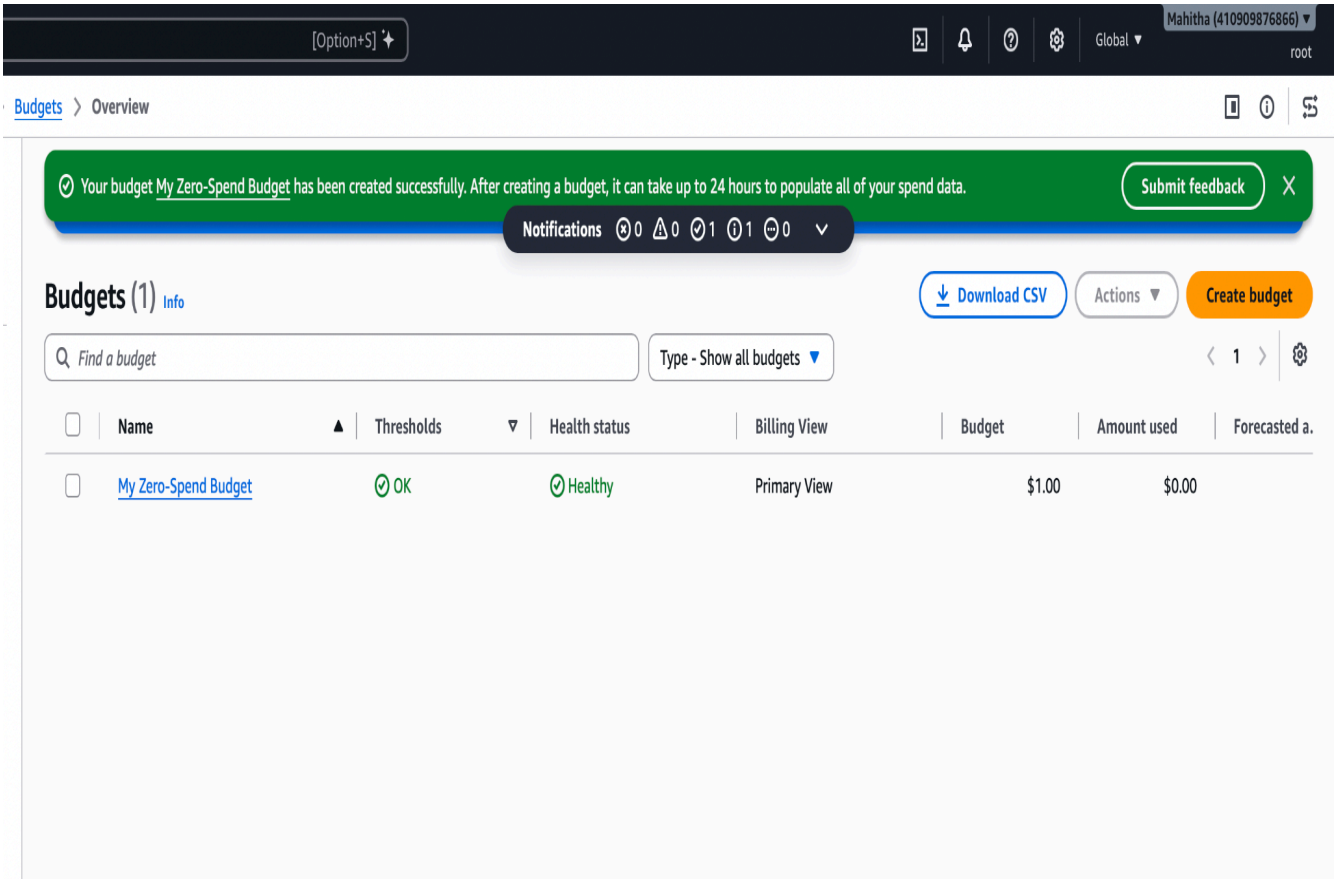

Console access enabled.
Summary
ARN
arn:aws:iam::410909876866:user/mahitha
Console access
Enabled without MFA
Access key 1
Create access key
Created
November 10, 2026, 09:09 (UTC+05
Last console sign-in
Permissions
Groups
Console password
You have successfully enabled the user's new password.
This is the only time you can view this password. After you close this window, if the password is lost, you must create a new one.
Console sign-in URL
https://410909876866.signin.aws.amazon.com/console
User name
mahitha
Console password
{]l[W8G
Hide
Download .csv file
Close
Console sign-in
Manage console access
Console sign-in link
https://410909876866.signin.a
-10 09:10 GMT+5:30)
Multi-factor authenticati
Remove
Resync
Assign MFA device
Use MFA to increase the security of y
e. Each user can have a maximum of 8 MFA devices assigned. Learn more
Type
Identifier
Certifications
Created on
No MFA devices. Assign an MFA device to improve the security of your AWS environment
Assign MFA device
Access keys (0)
Create access key
Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. Learn more


aws
Search
[Option+S]
Global
Mahitha (410909876866)
root
IAM > IAM users > mahitha > Create access key
This is the only time that the secret access key can be viewed or downloaded. You cannot recover it later. However, you can create a new access key any time.
Step 1
Access key best practices & alternatives
Step 2 - optional
Set description tag
Step 3
Retrieve access keys
Retrieve access keys Info
Access key
If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.
Access key
Secret access key
AKIAV7LBDEKBD3QTNX6G
********** Show
Access key best practices
Never store your access key in plain text, in a code repository, or in code.
Disable or delete access key when no longer needed.
Enable least-privilege permissions.
Rotate access keys regularly.
For more details about managing access keys, see the best practices for managing AWS access keys.
Download .csv file
Done
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Search
[Option+S]
Global
Mahitha (410909876866)
root
IAM > IAM users > mahitha
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Policy simulator
Credential report
Multi-account access
Account access manager
Organization activity
Service control policies
Resource control policies
Related consoles
IAM Identity Center
mahitha Info
Delete
Summary
ARN
arn:aws:iam::410909876866:user/mahitha
Console access
Enabled without MFA
Access key 1
AKIAV7LBDEKBD3QTNX6G - Active
Never used. Created today.
Created
November 10, 2026, 09:09 (UTC+05:30)
Last console sign-in
Never
Access key 2
Create access key
Permissions
Groups
Tags
Security credentials
Last Accessed
Console sign-in
Manage console access
Console sign-in link
https://410909876866.signin.aws.amazon.com/console
Console password
Updated 3 minutes ago (2026-11-10 09:10 GMT+5:30)
Last console sign-in
Never
Multi-factor authentication (MFA) (0)
Remove
Resync
Assign MFA device
Use MFA to increase the security of your AWS environment. Signing in with MFA requires an authentication code from an MFA device. Each user can have a maximum of 8 MFA devices assigned. Learn more
Type
Identifier
Certifications
Created on
No MFA devices. Assign an MFA device to improve the security of your AWS environment
Assign MFA device
Access keys (1)
Create access key
Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. Learn more
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Cookie preferences


[Option+S]
Global
Mahitha (410909876866)
root
IAM users (2) Info
An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.
Delete user
Create user
Search
1
User name
Path
Groups
Last activity
MFA
Password age
Console last sign-in
Access key ID
Student
/
0
-
-
5 minutes
-
Active - AKIAV7LBDEK...
Student1
/
0
-
-
-
-
Active - AKIAV7LBDEK...


Billing and Cost Management > Budgets > Create budget
Choose budget type Info
Budget setup
Use a template (simplified)
Use the recommended configurations. You can change some configuration options after the budget is created.
Customize (advanced)
Customize a budget to set parameters specific to your use case. You can customize the time period, the start month, and specific accounts.
Billing View - optional
Select a billing view you want to create this data export for. This will not be editable after create. If no view is selected, this export will be created for your account data (primary view).
Templates - new
Choose a template that best matches your use case.
Zero spend budget
Create a budget that notifies you once your spending exceeds $0.01.
Monthly cost budget
Create a monthly budget that notifies you if you exceed, or are forecasted to exceed, the budget amount.
Daily Savings Plans coverage budget
Create a coverage budget for your Savings Plans that notifies you when you fall below the defined target.
Daily reservation utilization budget
Create a utilization budget for your reservations that notifies you when you fall below the defined target.
Zero spend budget - Template
Budget name
Provide a descriptive name for this budget.
My Zero-Spend Budget
Names must be between 1-100 characters.
Email recipients
Specify the email recipients you want to notify when the threshold has exceeded.
Separate email addresses using commas
Maximum number of email recipients is 10.


Budgets > Overview
Your budget My Zero-Spend Budget has been created successfully. After creating a budget, it can take up to 24 hours to populate all of your spend data.
Submit feedback
Notifications
Budgets (1) Info
Download CSV
Actions
Create budget
Find a budget
Type - Show all budgets
Name
Thresholds
Health status
Billing View
Budget
Amount used
Forecasted a.
My Zero-Spend Budget
OK
Healthy
Primary View
$1.00
$0.00

Budget